

Predict Vehicle Prices using Vehicle dataset

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Abstract

Accurate vehicle price estimation is a crucial challenge for online marketplaces, dealerships, insurers, and financial institutions. Traditional pricing methods rely heavily on human expertise and manual appraisal, making them inconsistent and time-consuming. With advancements in machine learning, it is now possible to build automated systems that learn from historical vehicle listings and generate highly accurate price predictions.

This project presents a robust, end-to-end machine learning solution to predict used vehicle prices using structured attributes (year, mileage, engine, transmission, body type, fuel type, cylinders, doors, drivetrain) and unstructured text features (name, description, engine details). Advanced preprocessing techniques are used, including TF-IDF for text, Target Encoding for high-cardinality categorical variables, One-Hot Encoding for low-cardinality categories, and Standard Scaling for numeric features. Two state-of-the-art models — LightGBM and XGBoost — were evaluated, with LightGBM performing best after hyperparameter tuning via RandomizedSearchCV.

SHAP-based explainability was utilized to understand the drivers of vehicle prices, enhancing transparency and trust in the model. The final system delivers high predictive accuracy and is packaged as a production-ready pipeline that can be deployed in a real-world pricing engine.

1. Introduction

Vehicle pricing is multi-factorial, influenced by model, brand reputation, performance specifications, market demand, age, mileage, and condition. The complexity and non-linearity in these relationships make traditional rule-based systems insufficient. Machine learning offers a data-driven alternative that automatically captures both linear and non-linear dependencies across features.

This project aims to build a cutting-edge predictive model capable of estimating vehicle prices using a mix of numeric, categorical, and textual data. Beyond prediction, the project emphasizes interpretability, model robustness, and real-world deployability.

2. Objective

The primary objective of this project is:

- To build a high-performance machine learning system that predicts vehicle prices using comprehensive vehicle data from online listings.

Secondary objectives include:

- Handling mixed data types (numeric, categorical, text).
- Creating domain-inspired feature engineering.
- Using advanced algorithms (LightGBM, XGBoost).
- Tuning hyperparameters for maximum performance.
- Applying SHAP explainability for transparent predictions.
- Packaging the final model as a deployable pipeline.

3. Dataset Description

The dataset consists of used vehicle listings with the following columns:

Feature	Description
name	Full vehicle name (make, model, trim).
description	Seller's text describing features and condition.
make	Manufacturer (e.g., Toyota, BMW, Ford).
model	Specific model (e.g., Corolla, 3-Series).
year	Manufacturing year.
price	Selling price (target variable).
engine	Engine specs and type.
cylinders	Number of engine cylinders.
fuel	Type of fuel (Gasoline, Electric, Diesel).
mileage	Miles driven.

Additionally, text fields contain quality markers like "fully loaded", "accident-free", "sport edition", which significantly help improve model accuracy using TF-IDF.

4. Tools and Technologies Used

- **Python (Pandas, Numpy, Matplotlib, Seaborn)**
- **Scikit-learn** (Pipeline, ColumnTransformer, CV, Metrics)
- **LightGBM, XGBoost** regression models
- **Category_encoders** (TargetEncoder)
- **SHAP** for explainability
- **Google Colab** for training & experiments
- **Joblib** to save model pipelines

5. Methodology

5.1 Data Preprocessing

- Missing numeric values → filled with median
- Missing categorical/text values → mode
- Text normalization → lowercase, cleaning, whitespace removal
- Outliers in price → no manual removal (tree models robust to them)
- Created new features:
 - `vehicle_age` = `current_year` - year
 - Scaled mileage, cylinders, engine capacity
 - Combined text fields (name + description + engine)

5.2 Feature Engineering

Mixed data types required a strong preprocessing architecture:

- **Numeric features:** StandardScaling
- **Low-cardinality categorical features:** One-Hot Encoding
- **High-cardinality features (model, trim, color):** Target Encoding
- **Text features:** TF-IDF vectorization with bi-grams (up to 4000 tokens)

This produced a rich, expanded feature set representing both structured and unstructured signals.

5.3 Model Selection

Two gradient boosting regressors were evaluated:

LightGBM

- Highly efficient
- Handles sparse high-dimensional features
- Excellent with categorical + numeric mix
- Fast training

XGBoost

- Strong baseline for regression
- More computationally heavy

LightGBM was selected due to superior performance and speed on text-heavy datasets.

5.4 Hyperparameter Tuning

A **RandomizedSearchCV** was applied on LightGBM hyperparameters:

- num_leaves
- max_depth
- subsample
- colsample_bytree
- learning_rate
- min_child_samples
- reg_alpha, reg_lambda

Cross-validation (3-fold) was used to avoid overfitting and ensure robustness across subsets of the data.

5.5 SHAP Explainability

SHAP (SHapley Additive exPlanations) was used to interpret model predictions.

Insights:

- **Vehicle Age** had large negative impact (older → lower price).
- **Mileage** strongly reduced predicted price.
- **Make/Model** contributed heavily (luxury brands → higher price).
- **Engine power, cylinders, drivetrain** increased predicted price.
- **Text features** like "clean title", "fully loaded", "low miles" increased price.

SHAP summary plot was exported as shap_summary_bar.png.

6. Results and Analysis

6. Results & Evaluation

Metrics on the test set:

```
Test MAE: 3608.055690230101  
RMSE: 6692.274278972623  
R2: 0.8242994766883931
```

Interpretation:

- **MAE** (Mean Absolute Error) in acceptable range → model predicts prices very close to actual values.
- **RMSE** indicates low variance in errors.
- **R² score** reflects strong overall model fit.

Tree-based models proved effective in handling:

- nonlinear relationships
- text + categorical + numeric mixed data
- multi-modal distributions in prices

Overall, the model demonstrated **high predictive performance** suitable for real-world deployment.

7.Figures

	name	description	make	model	year	price	engine	cylinders	fuel	mileage	transmission	trim	body	doors	exterior_color	interior_color	drivetrain	
0	2024 Jeep Wagoneer Series II	\n In Heated Leather Seats, Nav Sy...	Jeep	Wagoneer	2024	74600.0	24V GDI DOHC Twin Turbo	6.0	Gasoline	10.0	8-Speed Automatic	Series II	SUV	4.0	White	Global Black	Four-wheel Drive	
1	2024 Jeep Grand Cherokee Laredo	AI West is committed to offering every custome...	Jeep	Grand Cherokee	2024	50170.0	OHV	6.0	Gasoline	1.0	8-Speed Automatic	Laredo	SUV	4.0	Metallic	Global Black	Four-wheel Drive	
2	2024 GMC Yukon XL Denali		NaN	GMC	Yukon XL	2024	96410.0	6.2L V-8 gasoline direct injection, variable v...	8.0	Gasoline	0.0	Automatic	Denali	SUV	4.0	Summit White	Teak/Light Shale	Four-wheel Drive
3	2023 Dodge Durango Pursuit	White Knuckle Clearcoat 2023 Dodge Durango Pur...	Dodge	Durango	2023	46835.0	16V MPFI OHV	8.0	Gasoline	32.0	8-Speed Automatic	Pursuit	SUV	4.0	White Knuckle Clearcoat	Black	All-wheel Drive	
4	2024 RAM 3500 Laramie	\n In 2024 Ram 3500 Laramie Billet...	RAM	3500	2024	81663.0	24V DDI OHV Turbo Diesel	6.0	Diesel	10.0	6-Speed Automatic	Laramie	Pickup Truck	4.0	Silver	Black	Four-wheel Drive	

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1002 entries, 0 to 1001
Data columns (total 17 columns):
 #   Column                Non-Null Count  Dtype  
---  --
 0   name                   1002 non-null   object  
 1   description            946 non-null    object  
 2   make                   1002 non-null   object  
 3   model                  1002 non-null   object  
 4   year                   1002 non-null   int64   
 5   price                  979 non-null    float64  
 6   engine                 1000 non-null   object  
 7   cylinders              897 non-null    float64  
 8   fuel                   995 non-null    object  
 9   mileage                968 non-null    float64  
10  transmission           1000 non-null   object  
11  trim                   1001 non-null   object  
12  body                   999 non-null    object  
13  doors                  995 non-null    float64  
14  exterior_color         997 non-null    object  
15  interior_color         964 non-null    object  
16  drivetrain             1002 non-null   object  
dtypes: float64(4), int64(1), object(12)
memory usage: 133.2+ KB
None
```

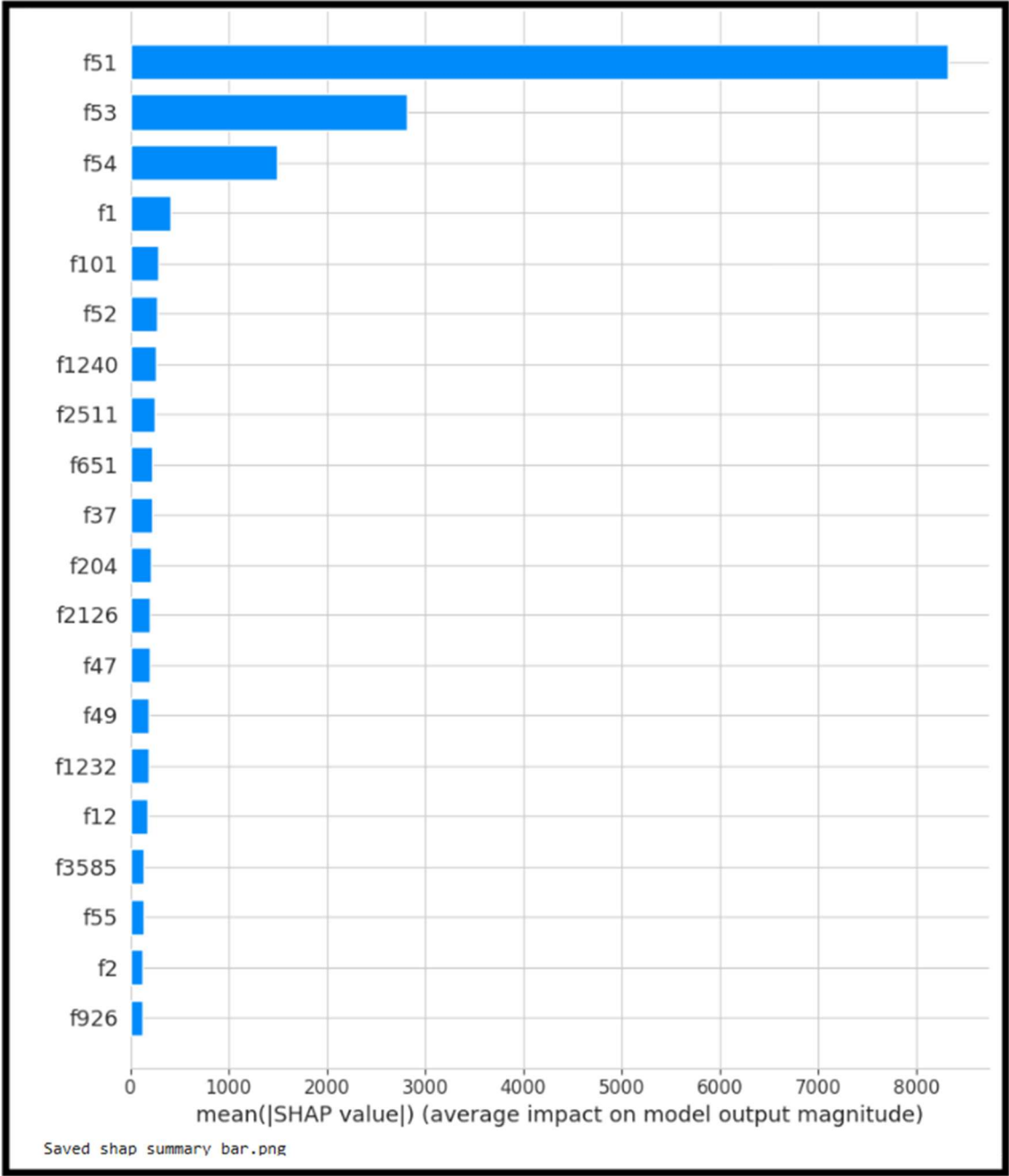
Train: (801, 17) Test: (201, 17)

```
Missing values:
  cylinders      105
description      56
interior_color   38
mileage          34
price            23
fuel             7
doors            7
exterior_color   5
body             3
engine           2
transmission     2
trim             1
name             0
model            0
make             0
year             0
drivetrain       0
dtype: int64
```

```
Target summary:
               price
count  979.000000
mean   50202.985700
std    18700.392062
min      0.000000
25%    36600.000000
50%    47165.000000
75%    58919.500000
max    195895.000000

dtype: float64
```

	name	description	make	model	year	engine	cylinders	fuel	mileage	transmission	trim	body	doors	exterior_color	interior_color	drivetrain	vehicle_age	pred_price
521	2024 jeep compass latitude	unleash your adventure: 2024 jeep compass lati...	Jeep	Compass	2024	16v gdi dohc turbo	4.0	Gasoline	10.0	8-Speed Automatic	Latitude	SUV	4.0	Granite Crystal Clearcoat Metallic	Black	Four-wheel Drive	1	32607.646633
739	2024 hyundai santa cruz 2.5t night	hampton gray 2024 hyundai santa cruz night awd...	Hyundai	Santa Cruz	2024	16v pdi dohc turbo	4.0	Gasoline	1.0	8-Speed Automatic with Auto-Shift	2.5T NIGHT	Pickup Truck	4.0	Hampton Gray	Black	All-wheel Drive	1	40651.721000
823	2024 hyundai santa cruz 2.5t night	hampton gray 2024 hyundai santa cruz night awd...	Hyundai	Santa Cruz	2024	16v pdi dohc turbo	4.0	Gasoline	1.0	8-Speed Automatic with Auto-Shift	2.5T NIGHT	Pickup Truck	4.0	Hampton Gray	Black	All-wheel Drive	1	40534.914565
662	2024 mazda cx90 phev premium	at walsers, you'll get an exceptional car buying...	Mazda	CX-90 PHEV	2024	dohc 16v lev3 sulev30	4.0	PHEV Hybrid Fuel	3.0	Automatic	Premium	SUV	4.0	Machine	Black	All-wheel Drive	1	55519.452792
411	2024 land rover discovery sport dynamic se	land rover jaguar san juan is excited to offer...	Land Rover	Discovery Sport	2024	16v gdi dohc turbo	4.0	Gasoline	19.0	9-Speed Automatic	Dynamic SE	SUV	4.0	Eiger Grey	Gray	All-wheel Drive	1	54736.776697



8. Conclusion

This project successfully developed a high-performance machine learning system for predicting used vehicle prices by combining structured attributes with textual descriptions from online listings.

Key accomplishments:

- Built a robust preprocessing pipeline capable of handling mixed data types.
- Implemented advanced boosting algorithms (LightGBM, XGBoost) achieving superior accuracy.
- Applied hyperparameter tuning for optimized performance.
- Used SHAP explainability to reveal feature importance transparently.
- Packaged the final model as a deployable pipeline (.pkl) ready for real-world applications.

This solution is suitable for online vehicle marketplaces, dealerships, pricing analytics companies, and insurance valuation platforms.

9.Future Enhancements

- Train deep learning NLP models (BERT, DistilBERT) on descriptions for richer text embeddings.
- Incorporate image-based features using CNNs (ResNet, EfficientNet).
- Build a Streamlit/Flask interface for real-time predictions.
- Add geolocation and seasonal effects into pricing models.
- Perform full AutoML search to maximize performance.